Question	Answer	Marks
6(a)(i)	lamp needs ‘high’ power/‘large’ current/‘large’ voltage	**B1**
	op-amp can deliver only a small current/small voltage	**B1**
6(a)(ii)	correct symbol for relay coil connected between output and earth	**B1**
	switch between mains supply and lamp	**B1**
6(b)(i)	vary light intensity at which lamp is switched on/off	**B1**
6(b)(ii)	so that relay operates for only one current/voltage direction **or** so that relay/lamp operates for either dark or light conditions	**B1**
6(c)	when light level increases, LDR resistance decreases	**B1**
	(R_{LDR} low,) so $V^{-} > V^{+}$, so V_{OUT} negative/–5 V (must be consistent with B1 mark)	**M1**
	or	
	when light level decreases, LDR resistance increases	**(B1)**
	(R_{LDR} high,) so $V^{-} < V^{+}$, so V_{OUT} is positive/+5 V (must be consistent with B1 mark)	**(M1)**
	lamp comes on as light level decreases **or** lamp goes off as light level increases	**A1**